

Metaphysical Foundations, General Systems Theory, and Combinatorial Discrete Morse Theory over Fragile Networks

System Architecture Invariant Manifest

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1 Introduction and Ontological Modalities

Following the formalisms of Edward Zalta’s Abstract Object Theory and Situation Theory, we isolate three distinct modal world-lines within a single unified processing manifold:

1. **The Historical Immutable World:** Rigidly designated actualities (s_{actual}) frozen in the past.
2. **The Fictional Mutable World:** Abstract objects encoding properties inside an isolated situation domain (s'), completely open to counterfactual planning rewrites.
3. **The Indexical Real World:** The concrete evaluation layer where sovereign actions and executions occur in real time.

2 The Mesarovic–Takahara Systems Framework

For large-scale fleet deployments ($N = 50$), complete point-to-point state-space tracking becomes computationally intractable. We address this by formalizing the system under a Mesarovic–Takahara input-output abstract response mapping:

$$S : M \times U \rightarrow Y \tag{1}$$

where U is the Disturbance Set generated by the non-linear Navier-Stokes weather advection field, M is the Control Input Set of discrete structural

graph interventions executed by the Domain-Specific Language (DSL) controller, and Y is the System Outcome Space tracking the layered physical, communication, and computational graph networks $(\mathcal{G}_P, \mathcal{G}_C, \mathcal{G}_{\text{Comp}})$.

3 Combinatorial Discrete Morse Theory over the Communication Mesh

To compress the continuous configuration space without losing topological properties, we apply Robin Forman's Discrete Morse Theory to the 30-mile line-of-sight communication hypergraph $\mathcal{G}_C$.

3.1 The Non-Degenerate Discrete Morse Function

Let K be a simplicial complex representing the communication network topology. A real-valued function $f : K \rightarrow \mathbb{R}$ is a valid Discrete Morse Function if for every p -simplex $\alpha \in K$, its value is strictly greater than all but at most one of its faces, and strictly less than all but at most one of its co-faces.

For a 3-node triangle complex with vertices $\mathcal{V} = \{0, 1, 2\}$ and edges $\mathcal{E} = \{(0, 1), (1, 2), (0, 2)\}$, we define our non-degenerate function mapping values explicitly as:

$$f(0) = 1.0, \quad f(1) = 3.0, \quad f(2) = 5.0 \quad (2)$$

$$f(0, 1) = 2.0, \quad f(1, 2) = 4.0, \quad f(0, 2) = 6.0 \quad (3)$$

3.2 Discrete Gradient Field and V-Path Calculation

A discrete gradient vector field V on K is a collection of pairs (α^p, β^{p+1}) such that α is a face of β and $f(\alpha) \geq f(\beta)$. A simplex is critical if it is not paired with any other cell. The pairing inequality condition allows combinatorial collapses along directed V-paths.

Applying our non-degenerate function values to the boundary operators yields the exact gradient pairings:

- $f(0, 1) = 2.0 \leq f(1) = 3.0 \implies$ Pairing Vector: $(1, (0, 1))$
- $f(1, 2) = 4.0 \leq f(2) = 5.0 \implies$ Pairing Vector: $(2, (1, 2))$

This collapses vertices 1 and 2 directly into their respective incident edges, leaving no further valid inequalities for the remaining cells.

3.3 Details of the Critical Simplices

A simplex is critical under the gradient field calculation if it remains entirely unpaired. Our non-degenerate topological extraction isolates the fundamental minimal topological invariants:

1. **Critical Minima Vertices [Traps]:** Vertex 0 is the only unpaired 0-cell ($f(0, 1) > f(0)$ and $f(0, 2) > f(0)$). It functions as the ultimate global attractor basin.
2. **Critical Saddle Edges [Tunnels]:** Edge (0, 2) remains completely unpaired ($f(0, 2) = 6.0 > f(2)$ and no higher co-faces exist). This isolates a stable 1-dimensional space-time cycle—the critical tunnel through which communication routing packets are scheduled.

3.4 Algebraic Schedule Words

Using the Symmetric Group S_3 acting as permutations over the network complex torso, task packet scheduling words are executed through the detected critical tunnel (0, 2). The sequential word composition $g_1 \cdot g_2 \cdot g_1$ routes task arrays recursively without computing statistical dynamic programming distributions, maintaining absolute algebraic mesh connectivity.